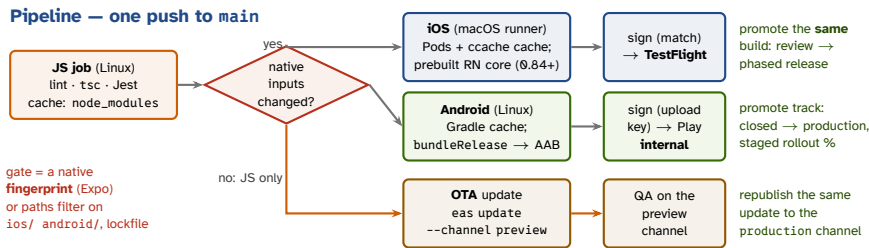
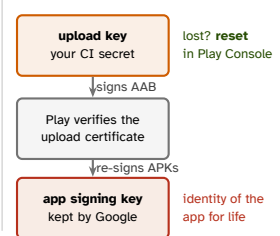


In one line: An RN pipeline is **two native builds** (iOS on macOS, Android on Linux) around one JS codebase, plus an **OTA lane** for JS-only changes. Make it fast with **lockfile-keyed caches**, reproducible with **signing secrets in the CI store** (Android keystore, iOS via match/API key), and safe by **promoting the tested artifact** — never rebuilding for release. On Android you sign with an **upload key**; Google re-signs with the **app signing key** it keeps.

Pipeline — one push to main



Play App Signing



Build matrix and caching

- Run cheap **JS checks once** on Linux; build natives only when native inputs changed; E2E (Detox/Maestro) on a sim/emulator for main or labelled PRs. macOS minutes cost the most.
- Cache keys = lockfiles:** yarn.lock/package-lock.json (node_modules or the package cache), Podfile.lock (ios/Pods), Gradle wrapper + *.gradle* (gradle/actions/setup-gradle). A branch-only key goes stale.
- Compile less:** prebuilt RN core .xcframeworks by default since 0.84; ccache (Podfile :ccache_enabled => true); Gradle configuration cache (0.79+); one ABI (reactNativeArchitectures=arm64-v8a) for debug/E2E only — store builds keep all.

Where it runs

EAS	Build (cloud or --local), Submit, Update, Workflows (.eas/workflows/*.yaml: build, submit, update, fingerprint jobs); manages keystore + iOS certs (eas credentials)
fastlane	Ruby lanes on any CI: gradle, upload_to_play_store (supply), match, gym, pilot — you own the machines
GitHub Actions / Bitrise	the runner + YAML/steps; call fastlane or EAS CLI from it

Android signing

- Play App Signing** is required for new apps (since Aug 2021, AAB): Google holds the app signing key; you sign uploads with the **upload key**. Lost/leaked upload key → **upload-key reset**; a self-held app key that is lost ends updates.
- Key: keytool -genkeypair -v -storetype PKCS12 -keyalg RSA -keysize 2048 -validity 10000; Google: validity ≥ 25 years (past 22 Oct 2033).
- The RN template signs **release with the debug keystore** until you add a signingConfigs.release. Passwords from ~/.gradle/gradle.properties (MYAPP_UPLOAD_STORE_PASSWORD...) or CI env ORG_GRADLE_PROJECT_<name>; keystore as a base64 secret decoded in the job.

iOS signing in CI

Same rules as native: **match** (readonly) or EAS-managed certs, an **App Store Connect API key** for upload, temp keychain per job — see signing-and-cicd.

QA builds before the store

iOS: TestFlight internal (no review) or ad hoc; Android: Play **internal testing** track (up to 100 testers) or Firebase App Distribution; EAS: *internal distribution* builds. Dev builds (expo-dev-client) are for engineers, never for QA sign-off.

Example — Android release job (GitHub Actions)

```
android:
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v4
    - uses: actions/setup-node@v4
      with: { node-version: 22, cache: yarn }
    - uses: actions/setup-java@v4
      with: { distribution: zulu, java-version: 17 }
    - uses: gradle/actions/setup-gradle@v4 # Gradle cache
    - run: yarn install --frozen-lockfile
    - run: echo "${KS}" | base64 -d > android/app/upload.keystore
    - run: { KS: "${ secrets.UPLOAD_KEYSTORE_B64 }" }
    - run: ./gradlew bundleRelease -PversionCode=${ github.run_number }
      working-directory: android
    - run: # read by signingConfigs.release as project properties
      ORG_GRADLE_PROJECT_MYAPP_UPLOAD_STORE_PASSWORD: ${ secrets.KS_PW }
      ORG_GRADLE_PROJECT_MYAPP_UPLOAD_KEY_PASSWORD: ${ secrets.KEY_PW }
```

Versions, OTA, promotion

- versionName/version is for humans; versionCode (integer) and iOS buildNumber must **rise on every upload** — derive from the CI run number, or EAS cli.appVersionSource: "remote" + autoIncrement: true (recommended since EAS CLI 12; it does not touch version).
- OTA:** each build profile carries a **channel**; an update reaches only builds with the same **runtimeVersion**. Test on preview, then eas update:republish --destination-channel production (same bundle); gradual: eas update --rollout-percentage 10, eas update:edit; undo: eas update:rollback.
- Stores:** TestFlight → review → phased release; Play internal → closed → production with a **staged rollout** you can halt.

Traps

- ! "Signed release" that is debug-signed — the template default.
- ! Two CI runs uploading the same versionCode: the second is rejected.
- ! A JS-only OTA that needs a new native module crashes old binaries — gate on runtimeVersion/fingerprint.
- ! Rebuilding for production after QA: you ship an untested binary.
- ! Secrets in EXP0_PUBLIC_* or .env end up in the bundle — CI secrets are for the *build*, not the app.

Remember

"Cache by lockfile, sign with secrets, bump by CI, promote the artifact."